NEWTON SCHOOL

PRE-CALCULUS | HOMEWORK

Recursion: Tower of Hanoi & Characteristic Equation

Student: Justin Koo **Date:** 2026-04-26 **Due:** 2026-05-03

Standards: *CCSS.MATH.CONTENT.HSF-IF.A.3 (recursive sequences) · HSF-BF.A.2 (translate recursive ↔ closed form). Today's focus: shift trick for non-homogeneous, and characteristic equation for homogeneous.*

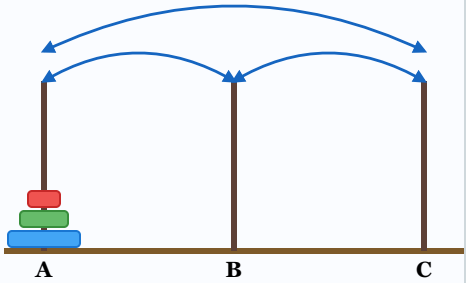
Problem 1 · Tower of Hanoi — Two Variants & the Shift Trick

📘 Setup — Compare the two variants

Both versions use 3 pegs **A — B — C** with n disks (largest at the bottom). Move one disk at a time, never a larger disk onto a smaller one. The **only** difference is which moves are *legal*:

Original Hanoi

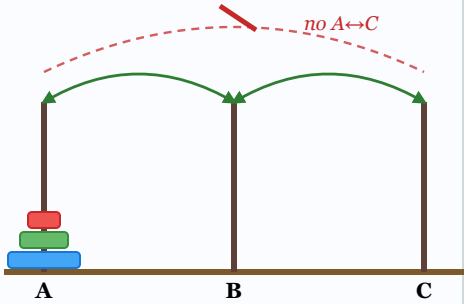
*disks may move **between any two pegs***



$$T_n = 2T_{n-1} + 1, \quad T_1 = 1$$

Adjacent-Only Hanoi

*disks move only **between adjacent pegs** (no A↔C)*



$$A_n = 3A_{n-1} + 2, \quad A_1 = 2$$

Why the recurrences differ: In *original* Hanoi, the largest disk moves once (A → C directly). In *adjacent-only*, it must hop A → B → C (two moves), and the smaller stack has to be moved out of the way **twice** (A→C, then C→A) before the big disk lands at C. That gives $3A_{n-1} + 2$ instead of $2T_{n-1} + 1$.

1a. Sketch a 3-disk move count. Using the diagrams above, fill in:

Original $T_3 = 2T_2 + 1 = 2($ _____ $) + 1 =$ _____

Adjacent $A_3 = 3A_2 + 2 = 3($ _____ $) + 2 =$ _____

(use $T_2 = 3$ and $A_2 = 8$ from the recurrences)

1b.

Solve A_n via the shift trick. The recurrence is $A_n = 3A_{n-1} + 2,\ A_1 = 2$.

Step 1. Substitute $U_n = A_n + c$. Plug into the recurrence and find the value of c that *kills the constant*.

Step 2. With your c, the recurrence on U_n becomes pure geometric. Compute U_1, then write U_n in closed form.

Step 3. Un-substitute $\rightarrow A_n =$ _____

Step 4. Verify A_2 and A_3 against your closed form.

1c.

New shift-trick exercise (different recurrence). A delivery scheduler has $Q_n = 4Q_{n-1} + 3$ with $Q_1 = 5$. Find the closed form using the same 4 steps.

Step 1. Find c in $U_n = Q_n + c$ that kills the constant.

Step 2. Solve the geometric U_n (compute U_1 first).

Step 3. Un-substitute $\rightarrow Q_n =$ _____

Step 4. Verify by computing Q_2 from the recurrence and from your closed form.

Problem 2 · Characteristic Equation Method

THEOREM ***Characteristic Equation for 2nd-Order Linear Homogeneous Recurrence***

For the recurrence $a_n = p \cdot a_{n-1} + q \cdot a_{n-2}$ with given a_0, a_1:

1. Write the **characteristic equation**: $x^2 - p\,x - q = 0$.
2. Solve for roots r_1, r_2 (use the quadratic formula or factor).
3. **Distinct roots** ($r_1 \neq r_2$): general solution is $a_n = C_1\, r_1^n + C_2\, r_2^n$.
4. Plug in a_0, a_1 to get a 2×2 system → solve for C_1, C_2.
5. Substitute back. The resulting closed form lets you compute a_n for any n **without** the recurrence.

Repeated root case ($r_1 = r_2 = r$): use $a_n = (C_1 + C_2 n) r^n$.

🧪 **WORKED EXAMPLE** ***Distinct integer roots — apply the 5-step method***

Solve $a_n = 3a_{n-1} - 2a_{n-2}$ with $a_0 = 1,\ a_1 = 3$.

Step 1. $p = 3,\ q = -2$. Characteristic equation: $x^2 - 3x + 2 = 0$.

Step 2. Factor: $(x-1)(x-2) = 0 \ \Rightarrow\ r_1 = 1,\ r_2 = 2$.

Step 3. General solution: $a_n = C_1(1)^n + C_2(2)^n = C_1 + C_2 \cdot 2^n$.

Step 4. Plug in a_0, a_1:

$a_0 = 1$: $C_1 + C_2 = 1$ ...(i)

$a_1 = 3$: $C_1 + 2C_2 = 3$ ...(ii)

Subtract (i) from (ii): $C_2 = 2 \ \Rightarrow\ C_1 = -1$.

Step 5. Closed form: $a_n = -1 + 2 \cdot 2^n = 2^{n+1} - 1$.

$$\boxed{a_n = 2^{n+1} - 1}$$

Sanity check: a_2 from recurrence: $3(3) - 2(1) = 7$. Closed form: $2^3 - 1 = 7$ ✓

→ Now apply the same 5 steps in 2a and 2b below. Show all your work.

2a. Apply to Lego/Fibonacci-shifted recurrence. Solve $L_n = L_{n-1} + L_{n-2}$ with $L_0 = L_1 = 1$.

Step 1. Identify p, q. Write the characteristic equation.

Step 2. Solve. Roots (call them φ and ψ):

Step 3. Write the general solution in **three equivalent forms**:

Symbolic form: $L_n = C_1\,\varphi^n + C_2\,\psi^n$ (using φ, ψ as names for the roots)

Radical form: $L_n = C_1\left(\frac{1+\sqrt{5}}{2}\right)^n + C_2\left(\frac{1-\sqrt{5}}{2}\right)^n$

Decimal (for sanity checks): $L_n \approx C_1(1.6180)^n + C_2(-0.6180)^n$

Step 4. Solve for C_1, C_2 using $L_0 = 1$ and $L_1 = 1$.

Step 5. Final closed form (Binet-shifted): $L_n =$

_____

Sanity check: compute L_5 from your closed form (use the decimal form). Does it equal 8 (from recurrence build-up $1, 1, 2, 3, 5, 8$)? _____

2b. New characteristic-equation exercise. Solve $a_n = 5\,a_{n-1} - 6\,a_{n-2}$ with $a_0 = 0,\ a_1 = 1$.

Step 1. Identify p, q and write the characteristic equation.

Step 2. Factor the quadratic (it factors over the integers — find the two roots).

Step 3. General solution: $a_n =$ _____

Step 4. Use $a_0 = 0, a_1 = 1$ to solve for C_1, C_2.

Step 5. Closed form: $a_n =$ _____

Verify: compute a_2, a_3, a_4 from the recurrence and from your closed form. Match?

📋 Submission Checklist

- Show every step of the shift trick / 5-step method — partial credit for setup, full credit for clean derivations.
- **Box every final answer.**
- State which method you're using at the start of each problem.
- Always do the sanity check — small arithmetic catches big errors.

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